Modelling Effects of Dam Installations on Sockeye Salmon

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Table of Contents

Background & Knowledge Gap----------**3**
Methods----------**4**
Data Collection----------4
Parameter Estimation and Confirmation----------4
Data Simulation and Rationale----------6
Parameter Manipulation and External Parameter Selection----------7
Results of the estimation process (Graphs and tables)----------**8**
MSY Simulation Results----------8
Conclusion----------**12**
Bibliography----------**13**

Background & Knowledge Gap

Pacific Salmon, a keystone species, anchors an entire web of life in British Columbia. Oceanic predators such as seals, wolves, bears, and even their own - preying on juvenile salmon - (Schipper, 2023) depend on salmon as a main form of nutrition. If salmon were to disappear or decline significantly in abundance, the entire ecosystem would take a major hit (Schipper, 2023).

British Columbia relies on hydroelectric dams, with more than 1,900 regulated dams supporting the province's hydro-dominant electricity system (Ministry of Environment and Climate Change Strategy, 2025), generating a total of 43,000 gigawatt-hours. Supplying more than 1.6 million residential, commercial, and industrial customers (*Hydro Electric Generation System*, n.d.). Government reports (Government of Canada, 2024) from BC's Dam Safety Program and the Auditor General highlight both the scale of this infrastructure and ongoing oversight challenges. BC's hydroelectric development, exemplified by major projects such as Site C, plays a central role in provincial energy planning while raising ecological and Indigenous governance concerns. In general, dams are deeply embedded in BC's energy landscape, shaping regional power supply, regulatory responsibilities, and long-term environmental management. Dams serve a variety of purposes, including flood control, water quality management, sediment control, and hydropower generation. However, their widespread use, particularly in British Columbia for hydroelectric power, has significant ecological impacts.

Dams have a profound impact on salmon populations. Salmon are anadromous; they spawn in freshwater streams and rivers, then migrate to the ocean, depending on saltwater to live as adults, and then migrate back to their spawning grounds to reproduce (Pacific Salmon Foundation, 2025). These dams impede and disrupt salmon migration, preventing them from easily moving upstream or downstream due to increased temperatures and altering the natural flow of water (Fisheries, 2023). The influence of dams on salmon migration and spawning success might also significantly alter the Maximum Sustainable Yield (MSY), the highest amount of catch that can be sustainably harvested without depleting the population. In this paper, we will use data from the Pacific Salmon Explorer to simulate a hypothetical population and model the effects of dam installations on the salmon population in British Columbia and how these changes will affect the Maximum Sustainable Yield.

Methods

Data Collection

The data used in this project comes from the Pacific Salmon Explorer website. We chose the Anderson–Seton Early Summer sockeye dataset, which contains complete and reliable spawner–recruit observations. The dataset we downloaded contains many salmon species, multiple Conservation Units (CUs), and numerous missing or placeholder values. These heterogeneous records cannot be combined for recruitment-dynamics modelling because each CU has unique productivity, life history traits, and environmental responses. The Ricker model we are using must be fitted to a single, internally consistent population. For this reason, we filtered the dataset to retain only year, spawners, and recruits for one CU.

This cleaned dataset provides the essential inputs for stock–recruit analysis, including variables for the Ricker model and for assessing population trends over time. Using the simplified CU-specific dataset also ensures that the data processing will run smoothly without numerical errors. Most importantly, it produces a biologically meaningful foundation for interpreting density dependence, peak spawning abundance, and recruitment dynamics.

Therefore, this CU-specific and cleaned dataset was selected because it is the only subset of the broader data that satisfies the analytical requirements of the Ricker model and supports reliable parameter estimation.

Parameter Estimation and Confirmation

Parameter estimation began by applying the Ricker stock–recruitment model to the data.

$$R = \alpha * S * e^{-\beta S}$$

The model parameters α (productivity) and β (density dependence) were first obtained using a linearized version of the Ricker equation,

$$R/S = \alpha * e^{-\beta S}$$

Where the logarithm of recruits per spawner was regressed against spawner abundance.

$$\ln(R/S) = \ln(\alpha) - \beta S$$

This expresses a simple linear regression model where the intercept equals $\ln(\alpha)$, and the slope equals $-\beta$. These initial estimates were then refined through nonlinear least squares to fit the full nonlinear Ricker curve directly to the observed data. Logging the Ricker model serves an important statistical and interpretive role in our analysis. By expressing the model in terms of ln(R/S), we convert a multiplicative stock–recruit relationship into an additive one, which helps stabilize variance and makes departures from the model easier to diagnose. The log-transformed formulation also allows us to interpret α and β more intuitively: α describes the expected recruits per spawner at very low density (i.e., potential productivity), while β controls the rate at which density dependence reduces recruitment as spawner abundance increases. Fitting a straight line to ln(R/S) versus spawners, therefore, provides a clear visual and statistical check on whether the empirical data exhibit the linear trend implied by the Ricker assumption.

Parameter Confirmation

To confirm that the estimated parameters successfully describe the true spawner–recruit relationship, predicted recruitment values were generated using the fitted model and plotted alongside the empirical data. The resulting curve exhibits the expected dome-shaped pattern, accurately capturing both the increasing returns at low spawner abundance and the density-dependent decline at high spawner levels. The close alignment between observed and modelled recruitment demonstrates that the fitted α and β parameters provide a biologically realistic representation of the population's stock–recruit dynamics.

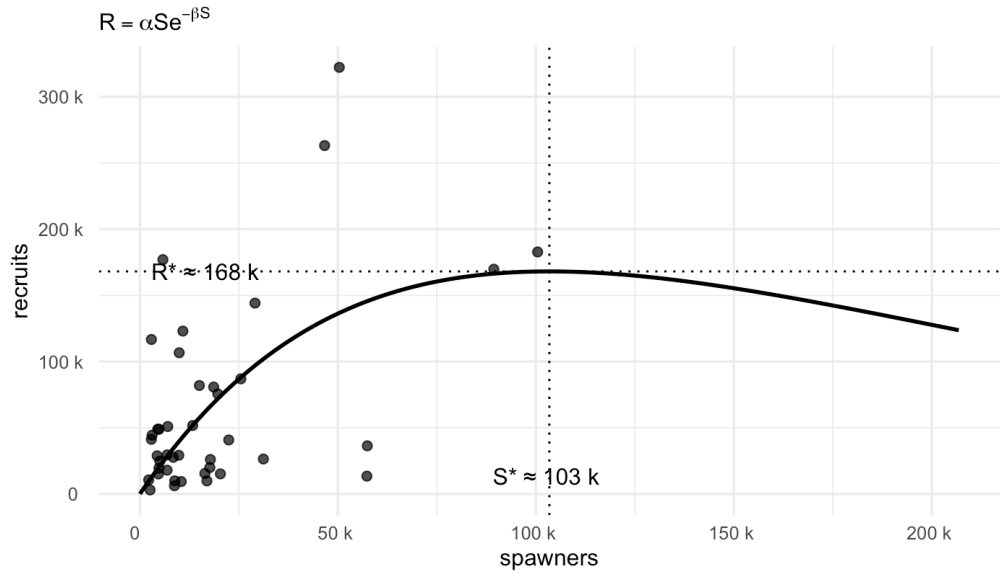


Figure 1: Ricker model using estimated α and β values generated from Pacific Salmon Explorer (2025)

These initial values were used as starting points for a nonlinear least-squares optimization, which fitted the full nonlinear Ricker function directly to the observed spawner–recruit data. After the final parameter estimates were obtained, a prediction grid of spawner values was constructed, extending beyond the observed range to capture the complete dome-shaped recruitment curve. The fitted Ricker function was then evaluated across this grid to generate smooth predicted recruitment values. Finally, observed data points and the predicted Ricker curve were plotted together, as shown in *Figure 1*, and key ecological quantities—such as the spawner level at maximum recruitment—were derived from the estimated parameters. This process results in a fully parameterized and visually interpretable Ricker stock–recruitment model for the selected sockeye population.

Data Simulation and Rationale

Following parameter estimation, simulated data were generated using the Ricker function to construct the density curve across a continuous range of spawner values. This simulation does not create new ecological observations but instead produces a theoretical recruitment curve representing the expected number of recruits at any spawner abundance,

based solely on the estimated α and β parameters. The purpose of generating this density curve is to understand long-term equilibrium behaviour and derive key population performance metrics such as maximum sustainable yield (MSY).

MSY requires determining the harvest level at which average recruitment balances fishing mortality, which cannot be obtained directly from raw observational data. Instead, the simulated density curve provides a continuous and smooth representation of the stock–recruit relationship, allowing the identification of maximum recruitment, harvest equilibria, and stability thresholds. Thus, simulation is essential for evaluating population resilience and sustainable harvest strategies under the Ricker model framework.

Parameter Manipulation and External Parameter Selection

The analysis also incorporates parameter manipulation to evaluate how external, non-population factors influence equilibrium dynamics and sustainable harvest levels. After constructing the simulated density curve, additional parameters—such as harvest rate, habitat degradation, and dam-related mortality—are systematically varied to assess their effects on equilibrium recruitment and yield. These external influences are incorporated through an extended Ricker formulation:

$$R = \alpha * s^{n} * S * e^{-\beta 1 s(1+\beta 2)}$$

where s^n represents cumulative survival across n dams and (1+β2)(1+β2) scales density dependence under habitat stress caused by the dams. According to Lawrence et al. (2016), the survival rate of passing one dam is independent of other passings of previous or subsequent dams. This is the reason why the survival rate was set as a constant and raised to the power of the number of dams. The effects on the density dependence were aggregated because we did not have any good reference for each effect. For the dam effects on the density dependence, we decided to aggregate the effects because we do not have any good reference for each effect. Manipulating these external parameters enables the model to examine scenarios not captured in the empirical dataset but essential for management decision-making—particularly for determining MSY and assessing population vulnerability under varying environmental pressures.

Results of the estimation process (Graphs and tables)

MSY Simulation Results

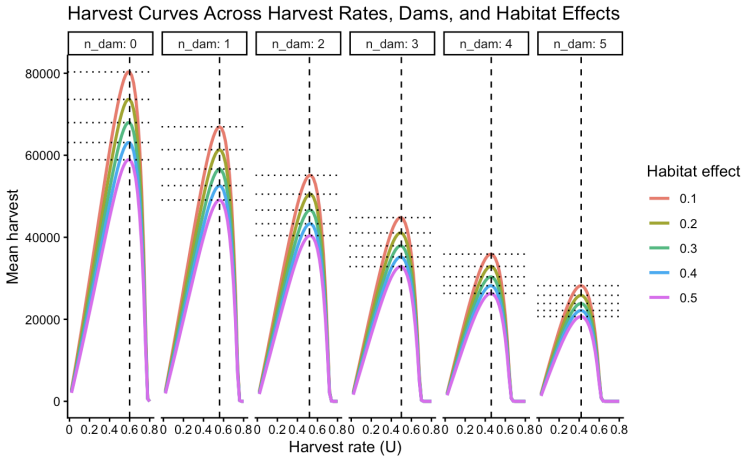


Figure 2

Figure 2 shows how simulated harvest outcomes vary with exploitation rate across different combinations of the number of dams and habitat degradation. In all scenarios, the harvest curves display the characteristic dome-shaped pattern of a Ricker-type stock, with yield rising with harvest rate until it reaches a single maximum and then collapsing sharply as exploitation becomes excessive. Increasing the number of dams produces a clear downward shift in the curves' heights. In the absence of dams, the stock supports a peak mean harvest exceeding 80,000 fish, but each additional dam reduces this value, cutting the attainable yield to less than half by the time 5 dams are present. This reflects the cumulative effect dams have on the mortality of salmon, where as the number of dams salmon have to pass through increases, it decreases the number of spawners successfully returning to the river. Thereby reduces the recruitment available for harvest in subsequent years. Habitat degradation intensifies this decline within every dam scenario. As the habitat-effect parameter increases from 0.1 to 0.5, the harvest curves become lower and narrower, indicating both reduced population productivity and stronger density dependence associated with diminished rearing and spawning capacity. The combined pattern demonstrates that degraded habitat and barriers to migration operate together to constrain sustainable harvest potential, shrinking both the magnitude of peak yield and the range of exploitation rates capable of maintaining long-term harvest. Figure 2, therefore, highlights the strong sensitivity of MSY-related outcomes to environmental and migratory pressures, underscoring the reduced resilience of the population under cumulative anthropogenic impacts.

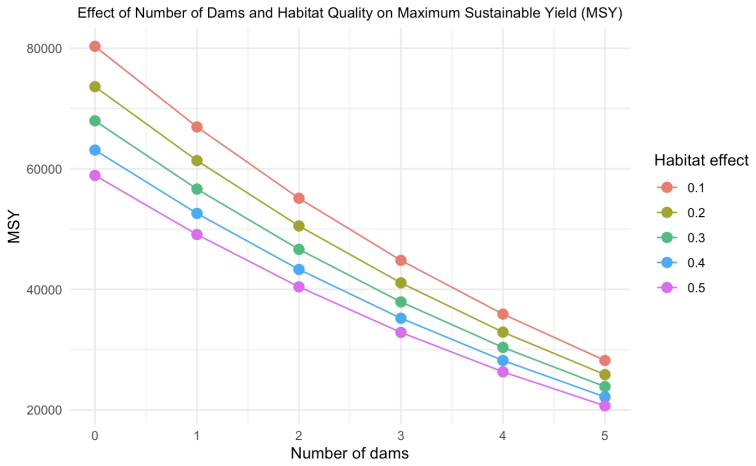


Figure 3

Figure 3 summarizes how maximum sustainable yield (MSY) varies across combinations of dam count and habitat degradation, providing a concise view of the cumulative constraints imposed by these factors on long-term harvest potential. Across all habitat conditions, MSY declines almost linearly as the number of dams increases, dropping from approximately 80,000 fish in the undammed scenario to values near 25,000 fish when five dams are present. This strong downward trend reflects the compounding migratory mortality embodied in the model, where each additional dam reduces the proportion of returning spawners, thereby diminishing the recruits available to sustain future harvests. Habitat degradation further amplifies this decline at every level of dam presence. As the habitat-effect parameter increases from 0.1 to 0.5, each MSY curve shifts downward, indicating reduced carrying capacity and heightened density dependence associated with poorer habitat conditions. Consequently, populations experiencing both multiple dams and degraded habitat exhibit the lowest MSY values, highlighting the interacting pressures that constrain population productivity. Taken together, the patterns in Figure 2 underscore that migration barriers and habitat loss do not operate in isolation; instead, they impose additive reductions in sustainable yield, limiting the range of conditions under which the stock can support robust long-term harvest.

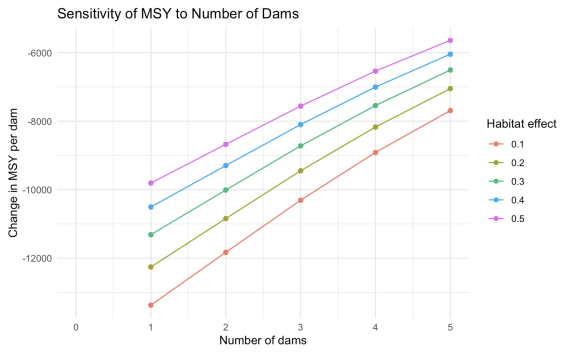


Figure 4

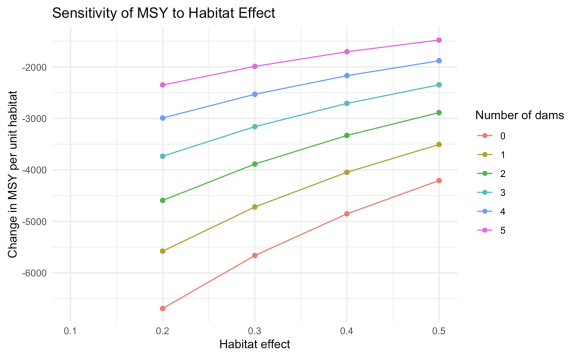


Figure 5

To gain a deeper understanding of how specific environmental factors affect Maximum Sustainable Yield (MSY), we plotted Figures 4 and 5. While Figure 3 provides an overall view of how MSY changes with the number of dams and habitat degradation, Figures 4 and 5 allow us to isolate and quantify the individual impacts of each factor. Figure 4 illustrates the change in MSY per additional dam, enabling us to understand how each new dam incrementally reduces MSY and reveals the cumulative effect of migratory barriers on salmon populations. Specifically, it shows how each dam lowers the potential harvest in an almost linear fashion, helping us quantify the direct loss in MSY associated with dam construction. Similarly, Figure 5 examines the change in MSY per unit of habitat degradation, offering insight into how varying levels of habitat quality influence the sustainable yield. It reveals how even small changes in habitat quality can significantly reduce the MSY, underscoring the importance of maintaining habitat integrity. These figures help provide a clearer view of the constraints on long-term fisheries sustainability in British Columbia.

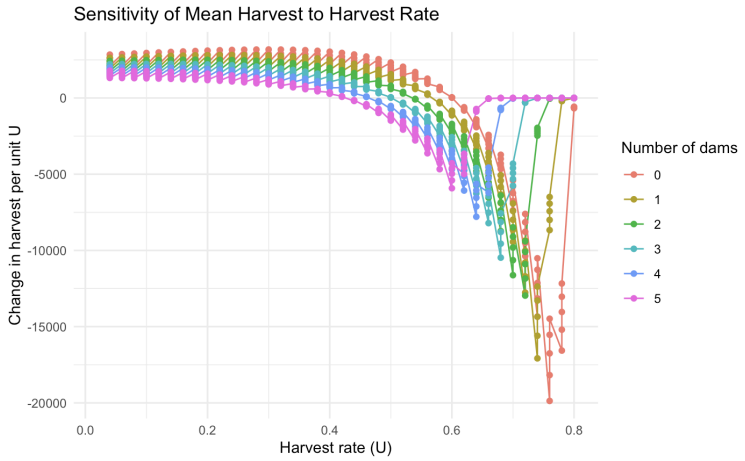


Figure 6

Additionally, we wanted to gain a clearer understanding of how harvest rates affect the mean harvest. Therefore, we decided to plot Figure 6, which examines the sensitivity of the mean harvest to harvest rate. While Figures 4 and 5 provide valuable insights into the individual effects of dams and habitat degradation on MSY, Figure 6 takes this analysis a step further by directly exploring how varying harvest rates influence the mean harvest across different dam scenarios. This plot gives us crucial information on the relationship between harvest rate and harvest outcomes, especially how much the harvest potential changes as the rate of harvest increases. It helps show how each additional dam affects the ability to harvest fish at varying rates, highlighting the sharp decline in harvest potential as the exploitation rate grows. As the number of dams increases, the sustainable harvest is more sensitive to changes in the harvest rate, causing much greater fluctuations in harvest potential compared to scenarios with fewer or no dams. The steep drop in mean harvest in the higher harvest rate ranges clearly demonstrates the loss of resilience in the salmon population as environmental stressors, such as dams, accumulate.

Ultimately, Figure 6 provides essential insights into how different harvest strategies, combined with the environmental impacts of dams, can influence the overall success of sustainable fisheries management. By understanding the sensitivity of mean harvest to harvest rate, we can make more informed decisions on how to balance exploitation and conservation, particularly in ecosystems where salmon populations are already under strain from anthropogenic pressures.

Conclusion

The findings of this project illustrate how strongly salmon population dynamics respond to cumulative ecological pressures, particularly those introduced by dams and habitat degradation. By applying and extending the Ricker stock–recruitment framework, we show that both migration barriers and declining habitat quality alter the fundamental relationship between spawners and recruits, lowering productivity, intensifying density dependence, and reducing the system's capacity to sustain harvest. Simulated scenarios reveal consistent declines in maximum recruitment and MSY as dam numbers increase or habitat conditions deteriorate, demonstrating how even moderate environmental stress can shift population equilibria and narrow the range of sustainable exploitation rates. These patterns confirm that long-term resilience is tightly linked to both migratory survival and habitat integrity. Overall, the analysis reinforces the importance of integrating habitat restoration, barrier mitigation, and conservative harvest strategies in fisheries management, especially in regions where salmon populations face multiple anthropogenic pressures.

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